AADK TASKS SESSION 4

Compose Basis And UI Layout

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TASK 1 – Composables Practice Reflection & Planning

1. Understanding Composable-Based UI Design

Jetpack Compose uses `@Composable` functions to define UI elements directly in Kotlin code. Instead of separating UI and logic using XML, Compose allows developers to describe UI layouts using composable functions that react automatically to state changes. This modern approach makes UI development more flexible, readable, and easier to maintain.

Through the "Practice: Compose Basics" codelab, I gained practical exposure to how composables are structured and combined to build user interfaces.

2. Practice Problem 1 – Vertical Layout Using Column

Problem Description:
This practice problem required creating a simple UI layout where multiple text elements were displayed vertically on the screen using a `Column` composable. The goal was to understand vertical arrangement and spacing between elements.

Planned Approach:
Before coding, I planned to use a `Column` composable with `Modifier.fillMaxSize()` to occupy the full screen. I decided to place multiple `Text` composables inside the column and apply padding and spacing using modifiers and `Spacer` to maintain proper layout structure.

Implementation & Reflection:
After implementation, the layout appeared clean and organized. Initially, adjusting spacing between text elements was slightly confusing, but using consistent padding values helped resolve the issue. This exercise helped me understand how vertical layouts are managed in Compose.

3. Practice Problem 2 – Horizontal Layout Using Row

Problem Description:
This problem focused on arranging UI elements horizontally using a `Row` composable. It involved placing an icon and text side by side while maintaining proper alignment.

Planned Approach:
I planned to use a `Row` composable with `verticalAlignment = Alignment.CenterVertically` so that both icon and text remain aligned. Padding modifiers were used to create spacing between the icon and the text.

Implementation & Reflection:
The implementation worked as expected, but I learned that modifier order plays a key role in alignment and spacing. This problem improved my understanding of horizontal layouts and alignment control in Compose.

4. Planning Checklist for New Composable UI Problems

Based on the practice exercises, I created a checklist for approaching new UI tasks:

- Understand the UI requirement clearly
- Sketch the layout on paper
- Identify required composables (Row, Column, Box, etc.)
- Decide alignment and spacing
- Apply modifiers for size and padding
- Test on different screen sizes
- Review accessibility and readability

5. Reflection

Planning before coding helped reduce confusion and made implementation smoother. Reflecting after implementation allowed me to identify mistakes and improve layout decisions. This approach improves efficiency and results in cleaner, more maintainable Compose UIs.

TASK 2 – Design & Document a Business Card Layout (Pre-Code)

1. Business Card Concept

The selected concept is a **Student Developer Business Card**, designed to display professional information in a clean and simple format. The business card includes the developer's name as the title, the role as Android Developer, a short description, and contact details such as phone number and email address.

This concept was chosen to maintain simplicity while presenting essential information clearly on a mobile screen.

2. Planned Screen Layout Description

The layout is planned in a vertical structure where elements are arranged from top to bottom. A profile icon is placed at the top, followed by the developer's name and role. Contact information is displayed at the bottom of the screen.

This layout ensures good readability and visual balance on small screens.

3. Composable Structure

The following composables are planned for the layout:

- `Column` for overall vertical structure
- `Icon` for displaying the profile image

- `Text` for name, role, and contact details
- `Row` for aligning labels and values in contact information

4. Spacing and Layout Planning

- 24dp padding around the entire layout
- 16dp space between the icon and name
- 8dp spacing between text elements
- Center alignment for main content

For larger screens or landscape orientation, padding will be increased to use available space effectively while keeping content centered.

5. Resource and Style Plan

Instead of using an external image file, an in-built Android icon is planned to simplify resource management.
Text styling is planned as follows:

- Name: 24sp, bold
- Role: 16sp, medium weight
- Contact text: 14sp

A light background with dark text ensures good contrast. Accessibility considerations include providing content descriptions for icons and maintaining sufficient spacing for touch interaction.

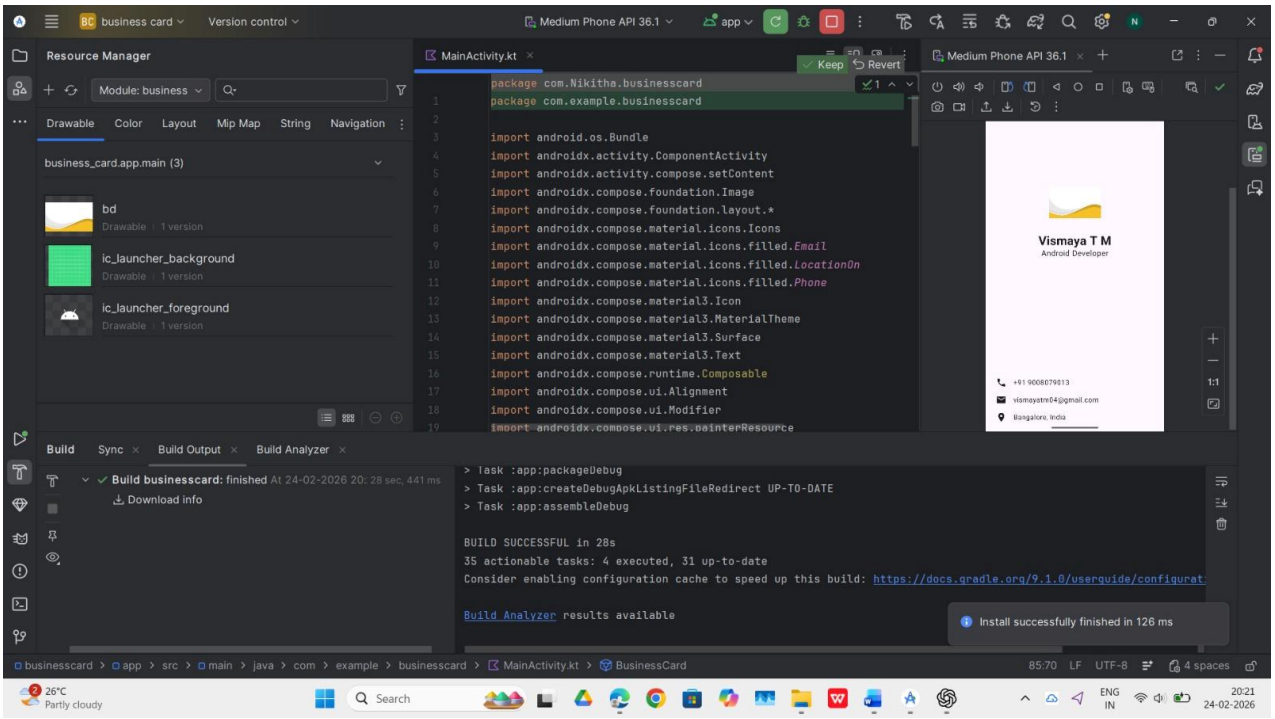
6. Reflection

Designing the layout before coding provided a clear blueprint for implementation. Planning composables, spacing, and resources reduced errors and saved development time. This approach makes UI development more structured and efficient.

TASK 3 – Layout Review & Improvement Plan of Business Card

1. Layout Review Overview

After implementing the business card layout using Jetpack Compose, the design was reviewed for alignment, readability, responsiveness, and accessibility. The layout was tested on different screen sizes using preview and emulator tools.



2. Visual Alignment and Spacing Review

The elements were properly centered and spaced consistently. Padding around the layout ensured that content did not touch screen edges. Overall alignment was visually balanced.

3. Readability and Resource Usage

Text sizes were readable, and contrast between text and background was adequate. Using an in-built icon instead of an external image helped reduce resource usage and avoided performance issues.

4. Responsiveness and Accessibility Review

The layout adapted well to different screen sizes due to the use of flexible composables. Accessibility was considered by adding content descriptions for icons and ensuring sufficient spacing for touch targets.

5. Improvement Plan

Issue Identified	Proposed Improvement	Priority
Subtitle text slightly small	Increase font size from 14sp to 16sp	Medium
Spacing on tablets	Increase padding to 32dp	Low

Issue Identified	Proposed Improvement	Priority
Visual emphasis	Add card background with rounded corners	Low

6. Reflection

Reviewing the layout helped me understand the importance of evaluating UI design beyond implementation. This process improved my awareness of accessibility, responsiveness, and visual quality. I will apply this review approach in future Compose projects to build better user interfaces.

Conclusion

Through these tasks, I gained a strong understanding of planning, designing, and reviewing Jetpack Compose layouts. I learned how to structure composables effectively, manage spacing and resources, and evaluate UI quality. This experience strengthened my confidence in Android UI development and helped me adopt a design-first mindset for building efficient and user-friendly applications.

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MindMatrix Internship - Android App Development using Gen AI

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